Hugo Marquerie 02/11/2025

Lem riesz

Lema 1 (de Riesz). *Sean Y, Z subespacios de un espacio vectorial normado X tales que $Y \subsetneqq Z$ e Y es cerrado* $\implies \forall \lambda \in (0,1) : \exists z \in Z : \|z\| = 1 \wedge \forall y \in Y : \|z - y\| \geqslant \lambda$.

***Demostración*:** Como $Y \subsetneqq Z$, podemos tomar $w \in Z \setminus Y$. Sea $a := \inf_{y \in Y} \|w - y\|$. Como Y es cerrado, Y^{c} es abierto y, por tanto, $\exists \varepsilon > 0 : a \geqslant \varepsilon$.

Sea $\lambda \in (0,1)$, podemos tomar $y_\lambda \in Y$ tal que $a \leqslant \|w - y_\lambda\| \leqslant \frac{a}{\lambda}$. Definimos $z := \frac{w - y_\lambda}{\|w - y_\lambda\|}$

$$\implies \forall y \in Y : \|z - y\| = \left\| \frac{w - y_\lambda}{\|w - y_\lambda\|} - y \right\| = \frac{1}{\|w - y_\lambda\|} \|w - y_\lambda - \|w - y_\lambda\| y\|$$
$$\geqslant \frac{a}{\|w - y_\lambda\|} \geqslant \frac{a}{a/\lambda} = \lambda.$$

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- `teo-bola-cerrada-compacta-imp-dim-finita`

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